

Sara Klaasen

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Education

ETH Zurich , Geophysics	Zurich, Switzerland
• Thesis distributed acoustic sensing for volcano monitoring • ETH medal	2020 – 2025
ETH Zurich , Geophysics	Zurich, Switzerland
	2018 – 2020
UCR , Liberal Arts and Sciences	Middelburg, Netherlands
• Magna cum Laude • Earth Sciences, biology, mathematics and physics	2014 – 2017

Experience

Research School of Earth Sciences, Australian National University , Postdoctoral researcher	Canberra, Australia
Researching DAS projects in New Zealand.	2026 – present
• Imaging Taupo Volcano.	1 year
ETH Zurich , Postdoctoral researcher	Zurich, Switzerland
	2025 – 2026
	1 year

Awards

ETH Medal	2026
The ETH Medal is given to outstanding doctoral theses.	
ETH Zurich	
ethz.ch/en/the-eth-zurich/education/awards/eth-medal.html	
OSPA	2024
Outstanding student presentation award	
Seismological Society of America Photonics Meeting	

Grants

SNF postdoc.mobility grant	2025
A grant to fund a 2-year postdoc position.	
Swiss National Science Foundation	
www.snf.ch/en/XIZpfY3iVS5KRRoD/funding/careers/postdoc-mobility	
VSBfonds grant	2018
A grant to support a Masters programme abroad.	
VSBfonds	
www.vsbfonds.nl	

Books

Fiber optic sensing for volcano monitoring and imaging volcanic processes

Jan 2025

Book chapter in Modern Volcano Monitoring

Jousset, Philippe and Currenti, Gilda and Murphy, Shane and Eibl, Eva and Caudron, Corentin and Retailleau, Lise and Wollin, Christopher and Klaasen, Sara and Nishimura, Takeshi and Fichtner, Andreas and Spica, Zack and Lemarchand, Arnaud and Krawczyk, Charlotte

doi.org/10.1007/978-3-031-86841-2_17

The leading edge of volcanology: Advances and innovations in monitoring and interpreting volcanic activity

Book chapter submitted to "The encyclopedia of Volcanoes, Third Edition"

Poland, Michael P and de Zeeuw-van Dalfsen, Elske and Aizawa, Koki and Klaasen, Sara and D'Arcy, Fiona and Nicholson, Emma J and Anderson, Kye

Publications

Submarine seismicity monitoring with distributed acoustic sensing near Santorini and Kolumbo Volcano

2026

Sara Klaasen, Thomas Hudson, Sanchit Sachdeva, Paraskevi Nomikou, Andreas Fichtner

doi.org/10.26443/seismica.v5i1.1845

Subglacial volcano monitoring with fibre-optic sensing: Grímsvötn, Iceland

2023

Sara Klaasen, Solvi Thrastarson, Yesim Cubuk-Sabuncu, Kristín Jónsdóttir, Lars Gebraad, Patrick Paitz, Andreas Fichtner

doi.org/10.30909/vol.06.02.301311

Distributed acoustic sensing in volcano-glacial environments — Mount Meager, British Columbia

2021

Sara Klaasen, Patrick Paitz, Nadja Lindner, Jan Dettmer, Andreas Fichtner

doi.org/10.1029/2021JB022358

Seismic constraints on glacier density

2025

Ariane Lanteri, Scott Keating, Lars Gebraad, Sara Klaasen, Marta Pienkowska-Cote, Olaf Eisen, Andrea Zunino, Kristín Jónsdóttir, Coen Hofstede, Dimitri Zigone, Andreas Fichtner

doi.org/10.1038/s41598-025-26440-z

The Seismogenic Thickness of Venus

2025

Julia Maia, Ana-Catalina Plesa, Iris van Zelst, Richard Ghail, Anna Gülcher, Mark Panning, Sven Peter Näsholm, Barbara de Toffoli, Anna Horleston, Krystyna Smolinski, Sara Klaasen, Robert Herrick, Raphael Garcia

doi.org/10.1038/s41598-025-26440-z

Towards a widely applicable earthquake detection algorithm for fibreoptic and hybrid fibreoptic-seismometer networks

2025

Thomas Hudson, Sara Klaasen, Olivier Fontaine, Conor Bacon, Kristin Jonsdottir, Andreas Fichtner

doi.org/10.1093/gji/ggae459

An illustrated guide to: Distributed and integrated fibre-optic sensing in seismology

2024

Andreas Fichtner, Fabian Walter, Patrick Paitz, Sara Klaasen, Daniel Bowden, Sebastian Noe, Nils Müller, Dominik Husmann, Jacques Morel

dx.doi.org/10.1016/j.eqs.2024.09.006

Seismic wave detectability on Venus using ground deformation sensors, infrasound sensors on balloons and airglow imagers	2024
Raphael Garcia, Iris van Zelst, Taichi Kawamura, Sven Peter Näsholm, Anna Horleston, Sara Klaasen, Maxence Lefevre, Celine Marie Solberg, Krystyna Smolinski, Ana-Catalina PLesa, Quentin Brissaud, Julia Maia, Simon Stähler, Philippe Lognonné, Mark Panning, Anna Gülcher, Richard Ghail, Barbara de Toffoli doi.org/10.1029/2024EA003670	
Modelling uncertainty in P-wave arrival-times retrieved from DAS data: case-studies from 15 fibre optic cables	2024
Emanuele Bozzi, Nicola Piana Agostinetti, Andreas Fichtner, Sara Klaasen, Arantza Ugalde, Biondo Biondi, S Yuan, Torsten Dahm, Marius Isken, Patrick Paitz, Fabian Walter, Alain Baird, Carlos Becerril, Takeshi Nishimura, Junzhu Shen, Tieyuan Zhu, Gilberto Saccorotti doi.org/10.1093/gji/ggae364	
Complex spatial distribution of onset amplitude and waveform correlation: case studies from different DAS experiments	2024
Emanuele Bozzi, Gilberto Saccorotti, Nicola Piana Agostinetti, Carlos Becerril, Andreas Fichtner, Sara Klaasen, Takeshi Nishimura, Junzhu Shen, Fabian Walter, Tieyuan Zhu doi.org/10.4430/bgo00458	
Challenges in submarine fiber-optic earthquake monitoring	2024
Jonas Igel, Sara Klaasen, Sebastian Noe, Paraskevi Nomikou, Konstantinos Korantzalos, Andreas Fichtner doi.org/10.1029/2024JB029556	
Fiber-optic observation of volcanic tremor through floating ice sheet resonance	2022
Andreas Fichtner, Sara Klaasen, Solvi Thrastarson, Yesim Cubuk-Sabuncu, Patrick Paitz, Kristin Jonsdottir doi.org/10.1785/0320220010	
Detecting Seismic Events with Computer Vision: Applications for Fiber-Optic Sensing	2021
Solvi Thrastarson, Robert Torfason, Sara Klaasen, Patrick Paitz, Yesim Cubuk-Sabuncu, Kristin Jonsdottir, Andreas Fichtner doi.org/10.1002/essoar.10509693.1	

Skills

Physics

Languages

Dutch

native

English

C2

German

B2

French

B1

Interests

Geophysics